

Smart quantum safe communication using AI and post quantum cryptography

Abstract

The rapid advancement of quantum computing creates a significant challenge for modern digital communication, as widely used public-key cryptographic techniques may become vulnerable to future quantum attacks. At the same time, conventional communication systems often lack intelligent mechanisms for continuously identifying abnormal user behavior and responding to emerging security threats. The main objective of this research is to develop a **Smart Quantum-Safe Communication System** that integrates **Post-Quantum Cryptography (PQC)** with **Artificial Intelligence (AI)** to provide secure, intelligent, and adaptive communication. This problem needs to be addressed because sensitive information transmitted through digital communication systems must be protected not only from current cyber threats but also from potential future quantum-based attacks. Existing approaches generally concentrate on post-quantum cryptographic protection or AI-based threat detection independently, creating a gap in developing an integrated and practical communication framework that provides both quantum-resistant protection and intelligent security monitoring. The proposed system uses **ML-KEM** for quantum-resistant key establishment, **AES-256-GCM** for secure message encryption, and **ML-DSA** for digital signatures and message authentication. In addition, an **Isolation Forest** model is used to analyze user activity, communication patterns, authentication attempts, and other security-related events to identify abnormal behavior and generate a risk score. Based on the detected risk level, the system provides an adaptive response, such as allowing normal communication, requesting re-authentication for suspicious activity, terminating a potentially compromised session, or blocking a high-risk user and generating a security alert. The proposed solution is designed as a **web-based secure communication system** with a backend service for authentication, session management, message processing, cryptographic operations, AI-based monitoring, and security control, while a lightweight database is used to securely maintain user information, communication records, security events, sessions, and AI-generated risk information. The significance of the proposed system lies in combining **quantum-resistant cryptographic protection with intelligent, behavior-based security monitoring** within a single communication framework. This approach provides an additional layer of adaptive protection and demonstrates how modern cryptographic techniques and artificial intelligence can work together to address both future quantum threats and present-day communication security challenges.

Keywords: Post-Quantum Cryptography, Artificial Intelligence, ML-KEM, ML-DSA, Secure Communication.

Problem Statement

The rapid development of quantum computing poses a potential threat to conventional public-key cryptographic techniques used to protect modern digital communication. At the same time, existing secure communication systems mainly focus on encryption and authentication, while providing limited capability for intelligent detection of abnormal user behavior. There is therefore a need for a communication framework that can provide **quantum-resistant protection** while also identifying suspicious activities in real time. The proposed project addresses this problem by integrating **Post-Quantum Cryptography (PQC)** with **Artificial Intelligence (AI)-based anomaly detection** in a secure web-based communication system. The system is designed to protect messages, authenticate users, monitor communication behavior, and dynamically respond to different levels of security risk.

Objectives

1. **To develop a quantum-safe communication mechanism** using ML-KEM for secure key establishment, AES-256-GCM for message encryption, and ML-DSA for digital signatures and message authentication.
2. **To develop an AI-based security monitoring mechanism** that analyzes user activities, authentication events, communication patterns, and other security-related behavior to detect abnormal or suspicious activities and generate an appropriate risk score.
3. **To implement an adaptive security framework** that responds to detected risk levels by allowing normal communication, requesting additional authentication, terminating suspicious sessions, or blocking high-risk users while maintaining relevant security records.

Domain / Research Area

The proposed research belongs to the following major domains:

- **Cybersecurity and Information Security**
- **Post-Quantum Cryptography**
- **Artificial Intelligence and Machine Learning**
- **Secure Communication Systems**
- **Network and Application Security**
- **Anomaly Detection and Adaptive Security**
- **Cryptographic Systems and Data Protection**

The project primarily focuses on the intersection of **Post-Quantum Cryptography, Artificial Intelligence, and Secure Digital Communication**.

Keywords

- Post-Quantum Cryptography (PQC)
- Artificial Intelligence
- Machine Learning
- ML-KEM
- ML-DSA
- AES-256-GCM
- Secure Communication
- Anomaly Detection
- Adaptive Security
- Quantum-Safe Communication

Software / Tools / Technologies Proposed

- **Python** – Main programming language used for backend development, cryptographic operations, AI-based anomaly detection, data processing, and security logic.
- **React.js** – Used to develop the interactive web-based user interface, including login, registration, secure chat, user status, and security-monitoring screens.
- **Node.js** – Used as a supporting runtime for the web application ecosystem, frontend development tools, package management, and real-time communication requirements where appropriate.
- **FastAPI** – Used as the primary backend framework for authentication, API services, session management, secure message processing, cryptographic operations, and communication between the frontend and security modules.
- **Firebase** – Used for cloud-based data and application services, including user-related information, communication records, security events, and application data management.
- **ML-KEM** – Used for post-quantum secure key establishment between communicating users, providing protection against potential future quantum attacks.
- **ML-DSA** – Used for post-quantum digital signatures to provide message authentication, integrity, and verification.
- **AES-256-GCM** – Used for efficient authenticated encryption of the actual communication messages after establishing a secure session key.
- **scikit-learn** – Used to implement the machine-learning component for detecting abnormal user and communication behavior.
- **Isolation Forest** – Used as the primary anomaly-detection algorithm to identify unusual activities such as repeated authentication failures, abnormal request rates, and unusual communication patterns.
- **Chart.js** – Used to create visual representations of AI risk scores, security events, user activity, and system security status in the administrator dashboard.
- **Postman** – Used for testing and validating backend APIs, authentication requests, communication services, and security-related endpoints.
- **Git / GitHub** – Used for source-code management, version control, collaboration, and maintaining different development versions of the project.

- **HTML, CSS and JavaScript** – Used as supporting web technologies for structuring, styling, and handling interactive elements of the web application.
- **WebSocket** – Can be used to provide real-time message delivery between users in the secure chat application.

Expected Outcome

The expected outcome of the project is a functional **Smart Quantum-Safe Communication System** that provides secure communication using post-quantum cryptographic techniques while continuously monitoring user behavior through an AI-based anomaly-detection mechanism. The system is expected to establish quantum-resistant communication sessions, protect messages through authenticated encryption, verify message authenticity, and identify suspicious activities based on behavioral patterns. According to the detected risk level, the system will provide adaptive security responses such as normal access, re-authentication, session termination, or user blocking. The final prototype is expected to demonstrate that combining **PQC-based communication with AI-driven security monitoring** can provide a more adaptive and future-oriented approach to protecting digital communication.

Literature Survey

Ref.	Authors	Year	Research Work	Methodology / Techniques	Key Contributions	Limitations / Research Gap
[1]	N. Lohmiller, S. Kaniewski, M. Menth, T. Heer	2025	A Survey of Post-Quantum Cryptography Migration in Vehicles	PQC migration analysis, cryptographic standards, V2V/V2X security, automotive communication	Reviews the transition from conventional cryptography to PQC in vehicular systems and identifies requirements and challenges for quantum-safe vehicle communication.	Focuses mainly on vehicular environments and migration strategies; does not integrate AI-based anomaly detection or adaptive security into a general-purpose secure communication application.
[2]	A. Sharma, S. Rani	2025	Post-Quantum Cryptography (PQC) for IoT-Consumer Electronics Devices Integrated With Deep Learning	PQC, hybrid cryptography, deep learning, anomaly detection, intrusion detection	Combines PQC with deep learning to improve security of connected consumer devices and investigates intelligent detection of suspicious activities.	Focuses on IoT/consumer electronics; does not provide a complete user-to-user secure communication platform using standardized ML-KEM and ML-DSA with adaptive risk-based responses.
[3]	H. Nguyen, S. Huda, Y.	2025	Security in Post-Quantum	Lattice-based cryptography, NTT,	Provides a comprehensive	Primarily a cryptographic

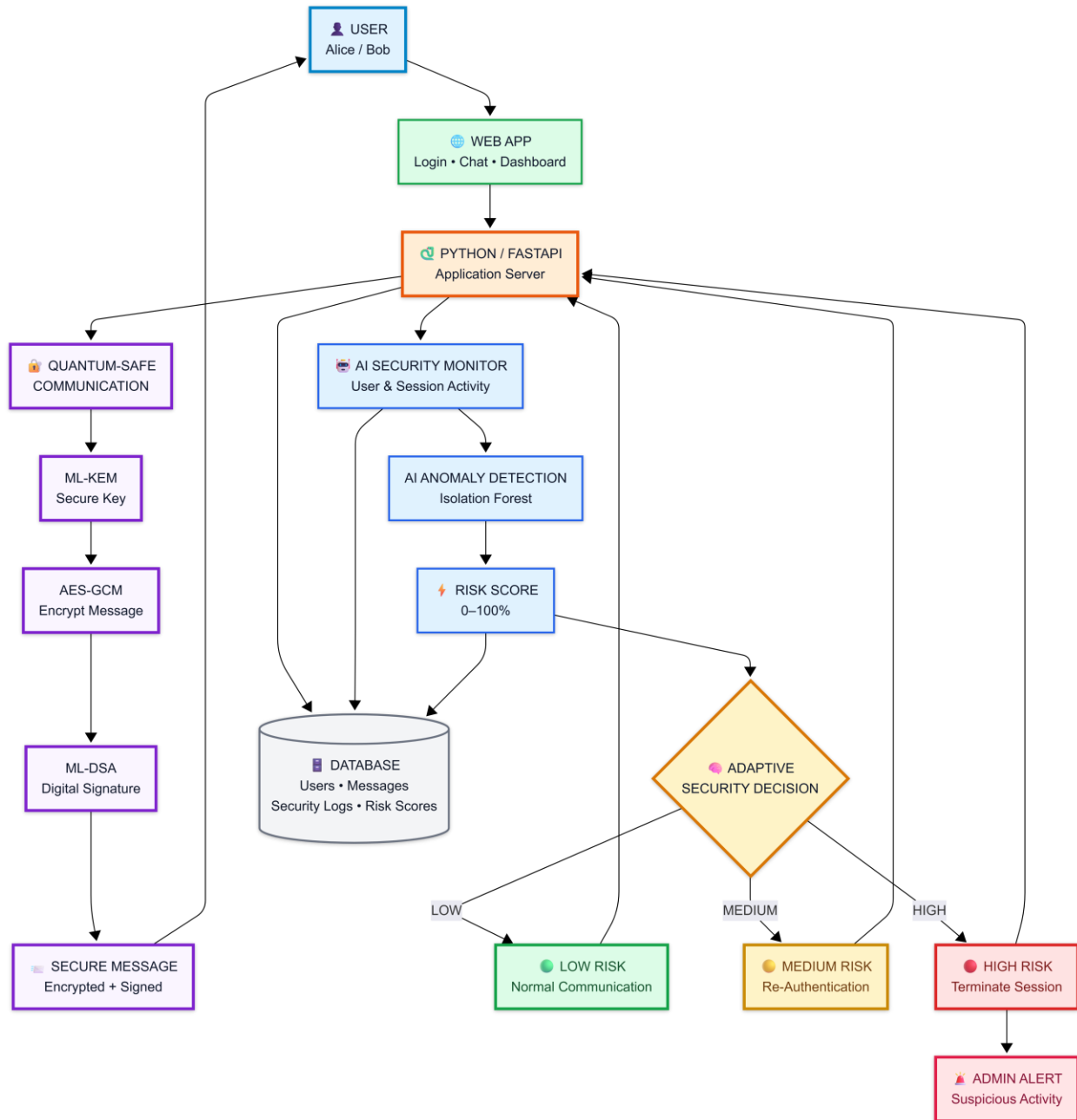
	Nogami, T. T. Nguyen		Era: A Comprehensive Survey on Lattice-Based Algorithms	PQC algorithms, digital signatures, hardware/software implementation	analysis of lattice-based PQC, implementation considerations, security characteristics, and optimization techniques.	survey; limited focus on integrating PQC with AI-based behavioral monitoring and real-time communication security.
[4]	Y. Wang, E. Shahril Ismail	2025	A Review on the Advances, Applications, and Future Prospects of Post-Quantum Cryptography in Blockchain and IoT	PQC review, blockchain security, IoT security, cryptographic standardization, hybrid approaches	Reviews PQC applications in blockchain and IoT and highlights standardization, deployment, and future research challenges.	Concentrates on blockchain and IoT applications rather than an AI-assisted communication system; adaptive user-level security is not addressed in detail.
[5]	B. Choudhury, A. Hota, M. Karmakar, S. Saha, A. Nag, S. Nandi	2025	A Comprehensive Survey on Pre Versus Post Quantum Security Schemes for 5G-Enabled IoT Applications	Comparison of classical and PQC schemes, 5G security, IoT security, lattice-based and code-based cryptography	Compares pre-quantum and post-quantum security approaches and identifies the requirements of quantum-safe protection for 5G-enabled IoT environments.	Focuses on 5G/IoT infrastructure and comparative analysis; does not implement an integrated AI-based threat detection and adaptive response mechanism.
[6]	K. B. A. Kumar, L. S. Mohith, K. Jain, P. Krishnan, N. Venkatachalam, R. Buyya	2025	Post-Quantum Cryptography-Based Multimedia Encryption Communication Scheme in IoT Consumer Electronics	PQC, NTRU, KEM, chaotic encryption, multimedia encryption, authentication	Proposes a PQC-based communication scheme for protecting multimedia data in connected consumer electronics and addresses confidentiality and authentication.	Primarily targets multimedia/IoT environments; the approach does not combine PQC communication with AI-based behavioral anomaly detection or dynamic security decisions.
[7]	S. Duraibi, A. Mujawib Alashjaee	2025	Lightweight Post-Quantum Secure Communication Protocol for IoT Devices Using Code-Based Cryptography	Code-based cryptography, lightweight PQC, ROLLO, mutual authentication, hash functions	Develops a lightweight quantum-resistant communication protocol intended for resource-constrained IoT	Uses code-based cryptography and focuses on IoT constraints; does not investigate AI-based anomaly detection or adaptive

					devices, emphasizing authentication and reduced computational requirements.	security policies for user communication.
[8]	M. A. Khan, F. Noor, S. Javaid, J. Żywiołek	2025	Implementation and Performance of Post-Quantum Cryptography for Resource Constrained Consumer Electronics	PQC implementation, performance evaluation, resource analysis, consumer electronics, communication overhead	Evaluates practical PQC implementation and performance on resource-constrained devices, showing the security and performance trade-offs associated with PQC deployment. (Springer)	Concentrates on implementation and performance of PQC rather than intelligent security monitoring; does not integrate AI-based anomaly detection or adaptive user-level responses.
[9]	A. Zafar, S. S. Iqbal	2025	Integrating Code-Based Post-Quantum Cryptography into SSL/TLS Protocols Through an Interoperable Hybrid Framework	Hybrid TLS, code-based PQC, McEliece, adaptive parameter selection, fallback negotiation, OpenSSL-OQS	Demonstrates a practical hybrid TLS framework with adaptive negotiation and evaluates latency, CPU usage, memory, and throughput for quantum-resistant communication. (Springer)	Focuses on TLS and code-based cryptography rather than application-level AI monitoring; does not provide behavioral risk analysis or AI-driven adaptive security decisions.
[10]	F. Barrett-danes, F. Ahmad	2025	Quantum Computing and Cybersecurity: A Rigorous Systematic Review of Emerging Threats, Post-Quantum Solutions, and Research Directions	Systematic literature review, PRISMA, PICo framework, quantum threat analysis, PQC and QKD evaluation	Provides a structured review of quantum threats, PQC solutions, implementation readiness, scalability, and research gaps across cybersecurity applications. (Springer)	Primarily provides a systematic research review rather than an implemented communication system; identifies the need for real-world testing but does not propose an AI-integrated adaptive communication framework.
[11]	B. Hanafi, M. Ali	2025	Analyzing the Research Impact in Post Quantum	Scientometric analysis, PRISMA, co-word analysis, thematic mapping,	Maps the development, major research themes,	Focuses on the research landscape rather than technical

			Cryptography Through Scientometric Evaluation	network analysis, Web of Science and Scopus	contributors, and emerging directions in PQC research and highlights gaps in scalability and real-world implementation. (Springer)	implementation; does not develop or evaluate a secure communication application combining PQC and AI.
[12]	J. A. Montenegro, R. Rios, J. Lopez-Cerezo	2026	A Performance Evaluation Framework for Post-Quantum TLS	OpenSSL-OQS, TLS, KEM benchmarking, PQC/hybrid configurations, handshake performance, network evaluation	Provides a framework for evaluating classical, hybrid, and PQC TLS configurations and analyzes handshake performance and communication overhead during migration to quantum-safe TLS.	Primarily evaluates TLS-level performance; does not include AI-based behavioral analysis, user risk scoring, or adaptive security responses within a communication application.
[13]	B. Ojetunde, T. Kurihara, K. Yano, T. Sakano, H. Yokoyama	2025	A Practical Implementation of Post-Quantum Cryptography for Secure Wireless Communication	Practical PQC implementation, KEM, secure wireless communication, AES, TLS/SSH-related security, performance evaluation	Demonstrates practical integration of PQC into secure wireless communication and evaluates the feasibility of quantum-resistant communication in realistic environments. (DOI)	Concentrates on wireless communication and cryptographic implementation; does not incorporate AI-based anomaly detection or adaptive security decisions based on user behavior.
[14]	R. Zhou, H. Guo, F. E. C. Teo	2026	Investigating Post-Quantum Cryptography to Secure Transmitted Data via Mobile Communication	PQC evaluation, Kyber, Dilithium, Falcon, SPHINCS+, mobile communication, 5G, experimental framework	Investigates PQC algorithms for protecting transmitted data in mobile communication and evaluates their suitability in 5G and future communication environments. (MDPI)	Focuses on mobile communication and cryptographic performance; does not integrate AI-driven anomaly detection, user behavior analysis, or adaptive security responses.

[15]	S.-Y. Chang, Q. Khan	2026	Post-Quantum Cryptography in Networking Protocols: Challenges, Solutions, and Future Directions	PQC networking, protocol integration, NIST standards, Internet/web/cellular networks, implementation analysis	Reviews PQC integration into networking protocols and identifies challenges related to scalability, interoperability, performance, and deployment across modern networks. (MDPI)	Provides a broad networking review but does not develop a complete AI-assisted secure communication platform with real-time anomaly detection and adaptive security control.
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Architecture diagram



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